

DEFINITIONS AND TERMINOLOGY

The following terms are provided to promote clarity and consistency in this disclosure. Unless otherwise indicated, the terms are not intended to be limiting, and variations, equivalents, and combinations are contemplated.

“Node” refers to a device or module configured to collect one or more physiologic or contextual signals and to provide such signals to a computing system for processing. A node may be wearable for extended measurement, session-based for baseline acquisition, or configured for physiologic context sensing.

“Intimate physiology” refers to physiologic signals, patterns, and responses associated with genital, pelvic, perineal, or intraluminal environments, and may include signals influenced by vascular dynamics, tissue mechanics, neuromuscular activity, hydration, temperature, and related properties.

“Longitudinal” refers to measurement across time such that trends, variability, recovery, and repeated patterns may be characterized across multiple time windows including minutes, hours, days, weeks, or months.

“Raw signal” refers to a sensor output prior to transformation into derived features or metrics. Raw signals may include voltage, resistance, capacitance, impedance, optical intensity, magnetic field magnitude, pressure readings, strain readings, inertial readings, and similar measurements.

“Feature” refers to a derived parameter computed from one or more raw signals. Features may be computed in real time or offline and may include filtered values, peaks, durations, slopes, variability measures, stability measures, frequency-domain measures, cross-sensor consistency measures, and other derived values.

“Metric” refers to an output computed from raw signals and/or features that is intended to represent a physiologic state, pattern, or change. Metrics may be numerical, categorical, indexed, or normalized, and may be presented as scores, zones, ranges, curves, or confidence indicators.

“Privacy-preserving output” refers to an output format that avoids explicit anatomical imaging or explicit anatomical labeling. Privacy-preserving outputs may include abstract graphs, bars, curves, indices, zones, trends, and confidence measures.

“Cross-device correlation” refers to a relationship between signals, features, or metrics derived from different nodes. Cross-device correlation may include temporal alignment, contextual conditioning, co-variation analysis, sequence detection, correlation scoring, lag analysis, and/or causality inference, and may be used to improve interpretation of intimate-region patterns by referencing physiologic context or baseline results.

“Cross-user correlation” refers to a relationship between privacy-preserving metrics of a first user and privacy-preserving metrics of a second user. Cross-user correlation may be performed across a range of pairing contexts, including male-to-female, male-to-male, female-to-female, and other pairings, and may be implemented without disclosing explicit anatomical imagery or explicit anatomical labels.

“Artifact” refers to signal distortion or confounding caused by motion, poor contact, misalignment, wear discontinuity, environmental changes, or other non-physiologic factors.

“Artifact rejection” refers to methods for reducing or removing artifacts, including by using inertial sensing, contact integrity sensing, multi-sensor consistency checks, motion gating, signal quality indices, spoof-resistance checks, and/or exclusion of low-confidence segments.

“Wear detection” refers to determining whether a node is being worn or in contact with tissue. Wear detection may be performed using contact sensors, impedance-based checks, capacitive proximity/contact confirmation, temperature gradients, optical coupling checks, pressure baselines, and combinations thereof.

“Baseline acquisition” refers to a routine performed to establish individualized reference parameters for a user. Baseline acquisition may include controlled contact, controlled radial stimulus, compression-response characterization, geometric measurement, tissue compliance measurement, endpoint sensing, and/or repeatable scan routines.

“Controlled radial stimulus” refers to controlled application of radial pressure or radial expansion to tissue, such as via an annular inflatable cuff, microbladder ring, constraint band, mechanical cam, micro-pump, manual pneumatic actuation, or combinations thereof, optionally measured by one or more pressure transducers.

“Compliance” refers to a measure of tissue or lumen response to applied stimulus, including deformation relative to force or pressure, and may be represented using indices, slopes, thresholds, hysteresis measures, stability measures, or other derived metrics.

“Geometry measurement” refers to measurement of physical dimensions or shape parameters using one or more sensing approaches including optical flow, time-of-flight, lidar, infrared endpoint sensing, pressure-based contact mapping, circumferential sensing, and combinations thereof.

“Anti-spoof” refers to methods configured to reduce the likelihood that a measurement is performed on a non-biologic object or in an invalid context. Anti-spoof methods may include physiologic confirmation via optical pulse sensing, contact pattern verification, impedance consistency checks, temperature dynamics checks, and/or multi-sensor agreement checks.

“Computing system” refers to one or more processors and memory configured to receive node data, perform signal processing, feature extraction, metric generation, correlation, and output presentation. The computing system may be implemented on a mobile device, wearable hub, desktop computer, server, cloud platform, or a combination thereof.